第十二章 大容量存储管理

1. None of the disk-scheduling disciplines, except FCFS, is truly fair (starvation may occur).

a. Explain why this assertion is true.

b. Describe a way to modify algorithms such as SCAN to ensure fairness.

c. Explain why fairness is an important goal in a time-sharing system.

d. Give examples of circumstances in which it is important that the operating system be unfair in serving I/O requests.

a. SSTF: Distant requests might never be selected.
SCAN: Requests arriving behind the head may repeated by be skipped until the header returns.

b. Introduce a deadline for each request. When a request has waited longer than its deadline, it must be serviced.

c. Prevents one users heavy I/O from delaying other users indefinitely.

d. ① Real-time tasks: Must have a strict deadline. so the requests must be served with priority.

② Paging: OS must prioritize page faults.

③ Journaling: File System journal commits must occur immediately for crash consistency.

2. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?

a. FCFS
b. SSTF
c. SCAN
d. LOOK
e. C-SCAN



a. FCFS: 143 → 86 → 1470 → 913 → 1774 → 948 → 1509 → 1022 → 1750 → 130

Distance: 57 + 1384 + 557 + 861 + 826 + 561 + 487 + 728 + 1620

= 7081

b. SSTF: 143 → 130 → 86 → 913 → 948 → 1022 → 1470 → 1509 → 1750 → 1774

Distance: 1745

c. SCAN: 143 → 913 → 948 → 1022 → 1470 → 1509 → 1750 → 1774 → 4999 → 130 → 86

Distance: 9769

d. Look: 143 → 913 → 948 → 1022 → 1470 → 1509 → 1750 → 1774 → 130 → 86

Distance: 3319

e. C-SCAN: 143 → 913 → 948 → 1022 → 1470 → 1509 → 1750 → 1774 → 4999 → 0 → 86 → 130

Distance: 9985